

Deliverable: Task TA-001 Requirement Notes

Project: House Price Prediction | AI/ML Project Sprint

Task ID: TA-001	Date: 03-Sep-2026
	Delivering: Requirement Notes

1. BUSINESS PROBLEM STATEMENT

In residential real estate, determining the true market value of a property is difficult for buyers and agencies. Traditional manual appraisals rely on personal judgment, causing significant operational challenges:

- Inconsistent Valuations:** Subjective assessments lead to overpricing (prolonged days on market) or underpricing (loss of revenue).
- Information Asymmetry:** Prospective buyers lack an objective, data-driven standard to evaluate whether a property's asking price is fair.
- Slow Valuation Cycles:** Manual comparative assessments take days to compile, delaying property sales.

2. BUSINESS OBJECTIVE & TARGET SOLUTION

The client requires an automated machine-learning pipeline to estimate fair market house prices based on physical and geographic property attributes.

- Machine Learning Objective:** Implement a supervised Linear Regression model to quantify the relationship between house attributes and closing transaction prices.
- Expected Business Outcome:** Generate real-time, unbiased pricing estimates to support decision-making and price transparency.

Technical Scope Clarification:
Linear Regression is strictly an algorithm for predicting continuous numerical values (selling price). It cannot predict personal identities such as the owner's name or buyer history. Those attributes represent relational database records, which will be maintained as metadata rather than model target variables.

3. PROBLEM FORMULATION & VARIABLE MAPPING

Component	Description	ML Role
Input Features (X)	Property attributes (Square footage, bedrooms, bathrooms, location, age, lot size).	Independent Variables
Target Variable (y)	Actual sale price of the house (SalePrice).	Dependent Variable (Continuous)
Model Type	Multiple Linear Regression ($y = \beta_0 + \sum \beta_i X_i + \epsilon$).	Supervised Regression Model
Success Metrics	R ² Score (Goodness of fit), RMSE, and MAE (Prediction error in currency).	Evaluation Criteria

4. END-TO-END PIPELINE WORKFLOW

- Data Ingestion:** Load the property dataset containing historical real estate sales.
- Quality Audit (TA-005):** Remove duplicate listings, handle missing values, and treat outliers.
- Feature Preparation (TA-003):** Encode categorical variables (e.g., neighborhood) and compute property age.
- Model Training:** Fit the Linear Regression model to find optimal feature coefficients (β_i).
- Evaluation & Delivery:** Evaluate performance against test benchmarks and output predictions for client use.